

RaceDay API Endpoint Plan

PROG6212 POE — Part 1, Section B

This table lists every API endpoint the RaceDay system will expose, matching the entities defined in the ERD (Section A) and the two system roles (Organiser, Participant). It will be implemented as-is in Part 2.

Authentication

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	/api/auth/register	Registers a new user as either an Organiser or a Participant.	None (public)	{ fullName, email, password, role }	201 Created — returns new user ID. 400 Bad Request — invalid data or role. 409 Conflict — email already registered.
POST	/api/auth/login	Authenticates a user and returns an access token.	None (public)	{ email, password }	200 OK — returns JWT token and user role. 401 Unauthorized — invalid credentials.

User Profile

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/users/me	Returns the profile of the currently logged-in user.	Any (logged in)	None	200 OK — returns user profile. 401 Unauthorized — no valid token.
PUT	/api/users/me	Updates the profile of the currently logged-in user.	Any (logged in)	{ fullName, email }	200 OK — profile updated. 400 Bad Request — invalid data. 401 Unauthorized.

Events

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/events	Lists all upcoming events, for participants to browse.	None (public)	None	200 OK — returns list of events.

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/events/{id}	Returns full detail for a single event.	None (public)	None	200 OK — returns event detail. 404 Not Found — event does not exist.
POST	/api/events	Creates a new event.	Organiser	{ eventName, eventDate, location, description }	201 Created — returns new event ID. 400 Bad Request — invalid data.
PUT	/api/events/{id}	Updates an existing event's details.	Organiser	{ eventName, eventDate, location, description }	200 OK — event updated. 403 Forbidden — not the owning organiser. 404 Not Found.
DELETE	/api/events/{id}	Deletes an event.	Organiser	None	204 No Content — event deleted. 403 Forbidden. 404 Not Found.

Categories

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/events/{eventId}/categories	Lists all categories available for a specific event.	None (public)	None	200 OK — returns list of categories. 404 Not Found — event does not exist.
POST	/api/events/{eventId}/categories	Adds a new category to an event.	Organiser	{ categoryName, distanceKm, price }	201 Created — returns new category ID. 403 Forbidden. 404 Not Found — event does not exist.

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
PUT	/api/categories/{id}	Updates an existing category.	Organiser	{ categoryName, distanceKm, price }	200 OK — category updated. 403 Forbidden. 404 Not Found.
DELETE	/api/categories/{id}	Deletes a category.	Organiser	None	204 No Content. 403 Forbidden. 404 Not Found.

Event Enrolments

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	/api/categories/{categoryId}/enrol	Enrols the logged-in participant into a category.	Participant	None (participant identified from token)	201 Created — returns new enrolment ID. 404 Not Found — category does not exist. 409 Conflict — already enrolled in this category.
GET	/api/users/me/enrolments	Returns all enrolments belonging to the logged-in participant.	Participant	None	200 OK — returns list of enrolments.
GET	/api/events/{eventId}/enrolments	Returns all participant enrolments for an event (for the	Organiser	None	200 OK — returns list of enrolments. 403 Forbidden — not the

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
		organiser to manage).			owning organiser. 404 Not Found.
DELETE	/api/enrolments/{id}	Cancels the logged-in participant's own enrolment.	Participant	None	204 No Content. 403 Forbidden — not the owning participant. 404 Not Found.

Results

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	/api/enrolments/{enrolmentId}/result	Captures or updates the result for a participant's enrolment.	Organiser	{ finishTime, position, status }	201 Created — result recorded. 403 Forbidden — not the owning organiser. 404 Not Found — enrolment does not exist.
GET	/api/users/me/results	Returns the logged-in participant's personal results history.	Participant	None	200 OK — returns list of results.
GET	/api/events/{eventId}/results	Returns the full results/leaderboard for an event.	None (public)	None	200 OK — returns list of

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
					results. 404 Not Found — event does not exist.

Notes on design decisions

- Routes are nested under their parent resource (e.g. `/api/events/{eventId}/categories`) where a category or enrolment only makes sense in the context of its parent — this keeps ownership explicit and matches the ERD's foreign key relationships.
- `POST /api/categories/{categoryId}/enrol` deliberately takes no request body beyond the authenticated user's identity, since the participant and category are already known from the token and the URL — this avoids letting a participant enrol *someone else*.
- role: Any (logged in) is used only for the profile endpoints, since both Organisers and Participants need to view/edit their own profile.
- Delete/update endpoints check ownership (403 Forbidden) before returning 404, so an organiser cannot delete another organiser's event or category.